RIS clusterProfiler wrapper

Marissa Macchietto

ClusterProfiler

- R package for analyzing and visualizing functional profiles (pathways) of gene and gene clusters

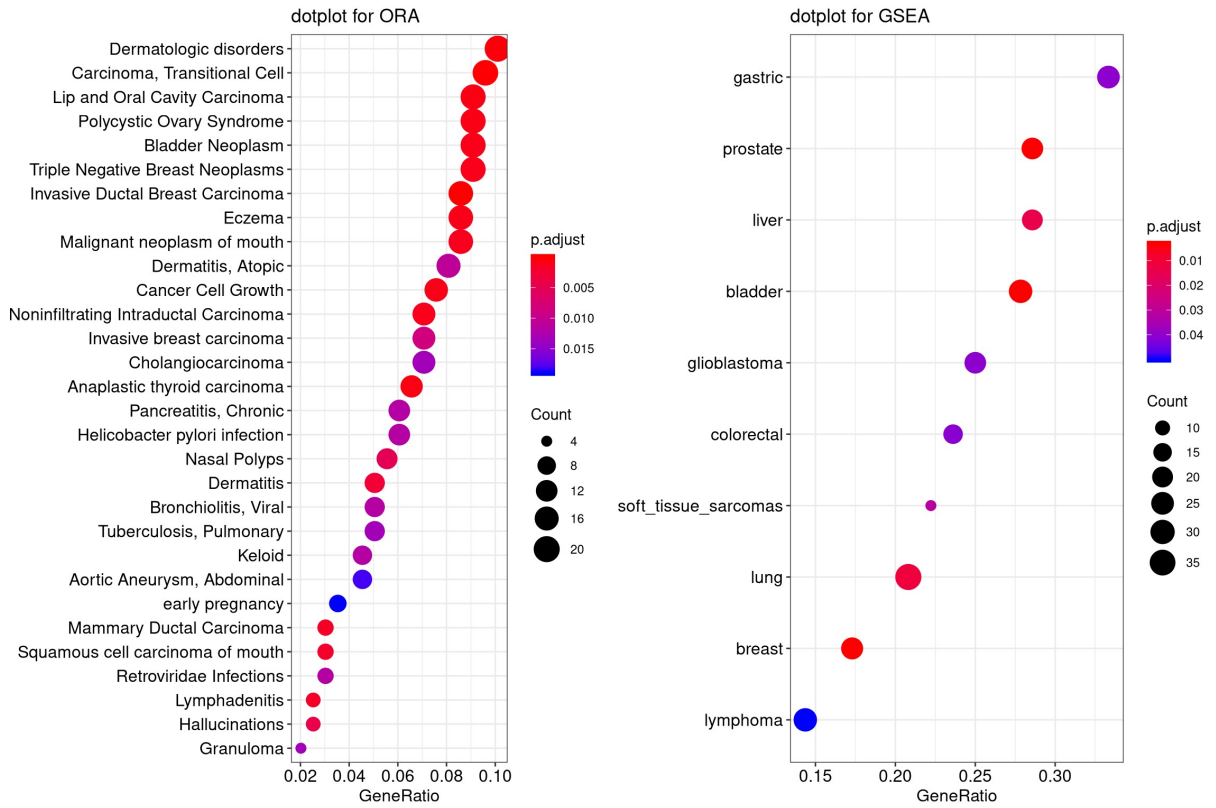
- **Supported analyses:**
  - Over-representation analysis
  - Gene set enrichment analysis
  - Biological theme comparison

- **Pathway databases:**
  - Gene Ontology (GO)
  - Kyoto Encyclopedia of Genes and Genomes (KEGG)
  - Reactome
  - MsigDB

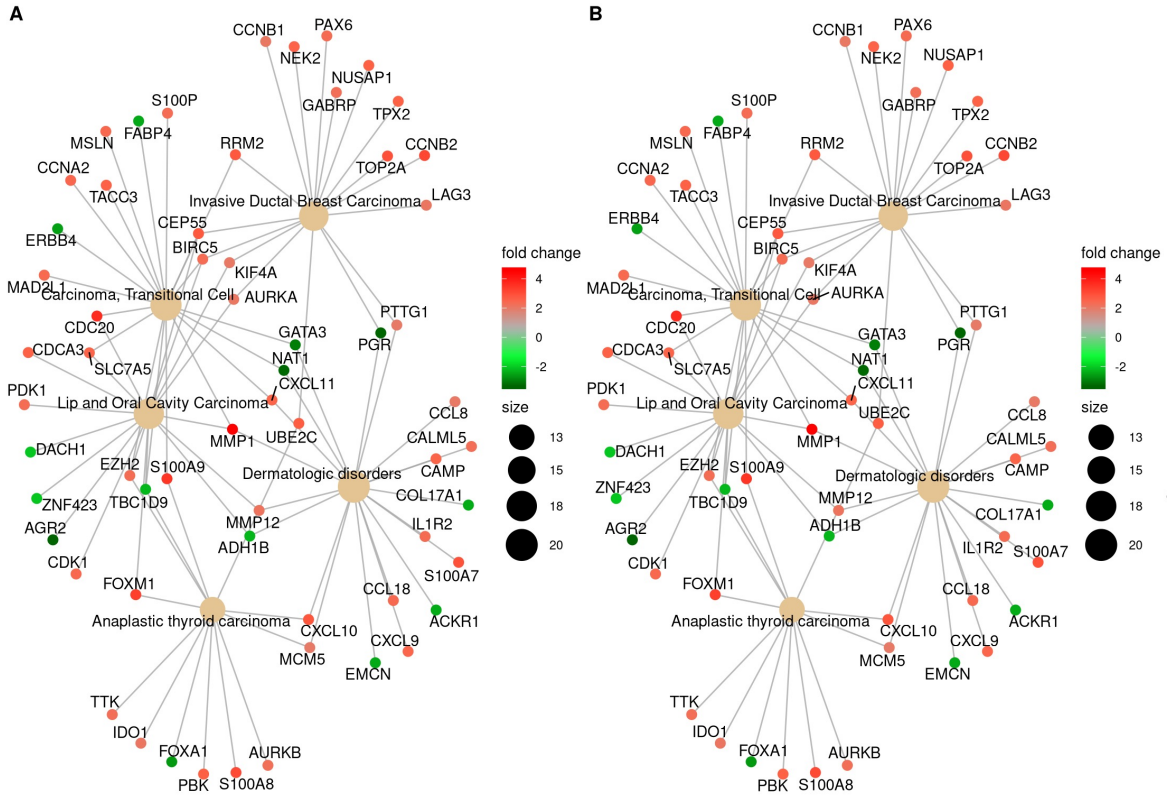
- **Visualization (plot types):**
  - barplots
  - cnetplot
  - dotplot
  - emapplot
  - gseaplot
  - goplot
  - upsetplot
  - heatplot
  - pathview plots (KEGG)
  - ridgeplot

ClusterProfiler – examples of analysis plots

dotplot



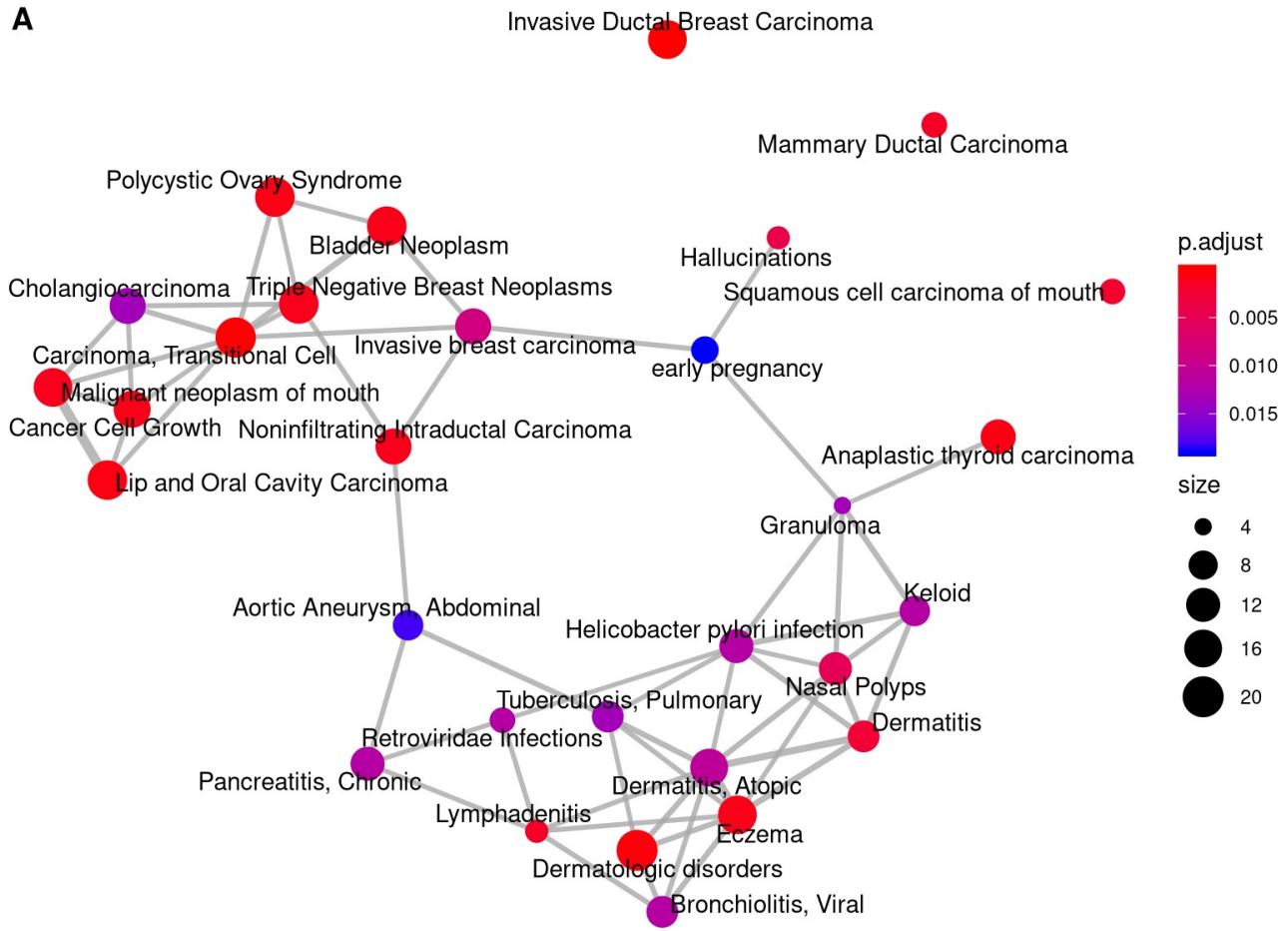
cnetplot



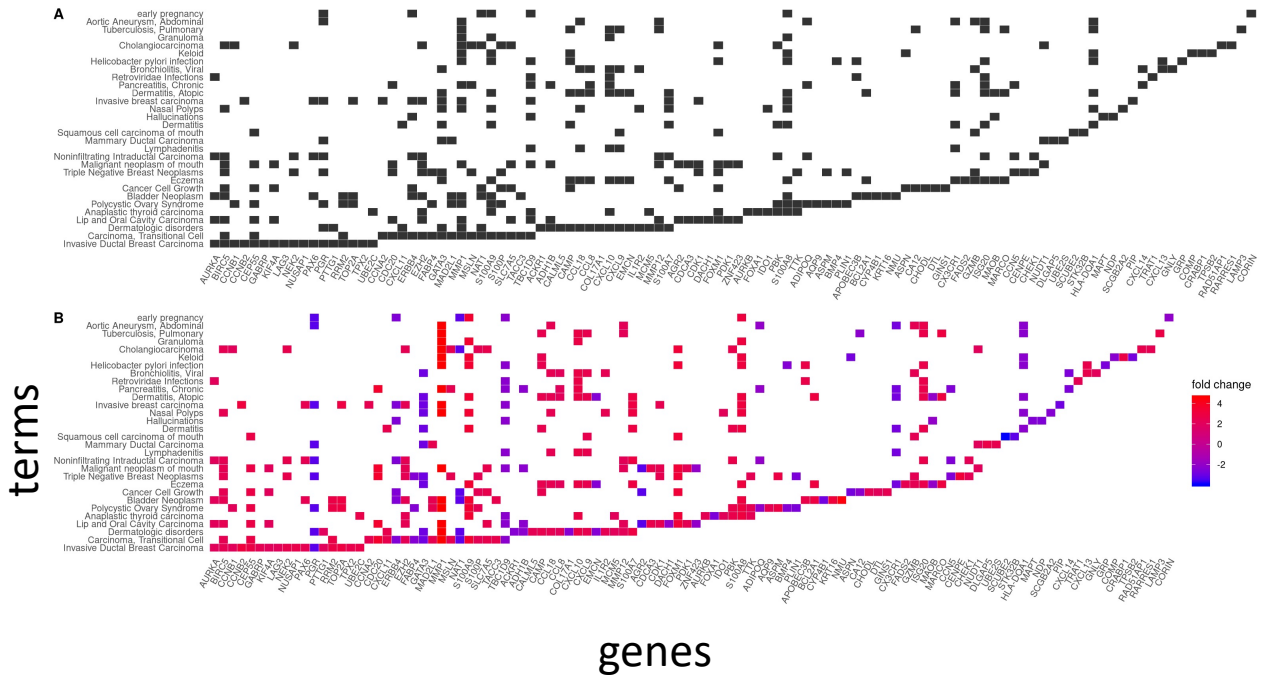
- All of these plots can be generated for GO, KEGG, Reactome, and others

ClusterProfiler – examples of analysis plots

emapplot

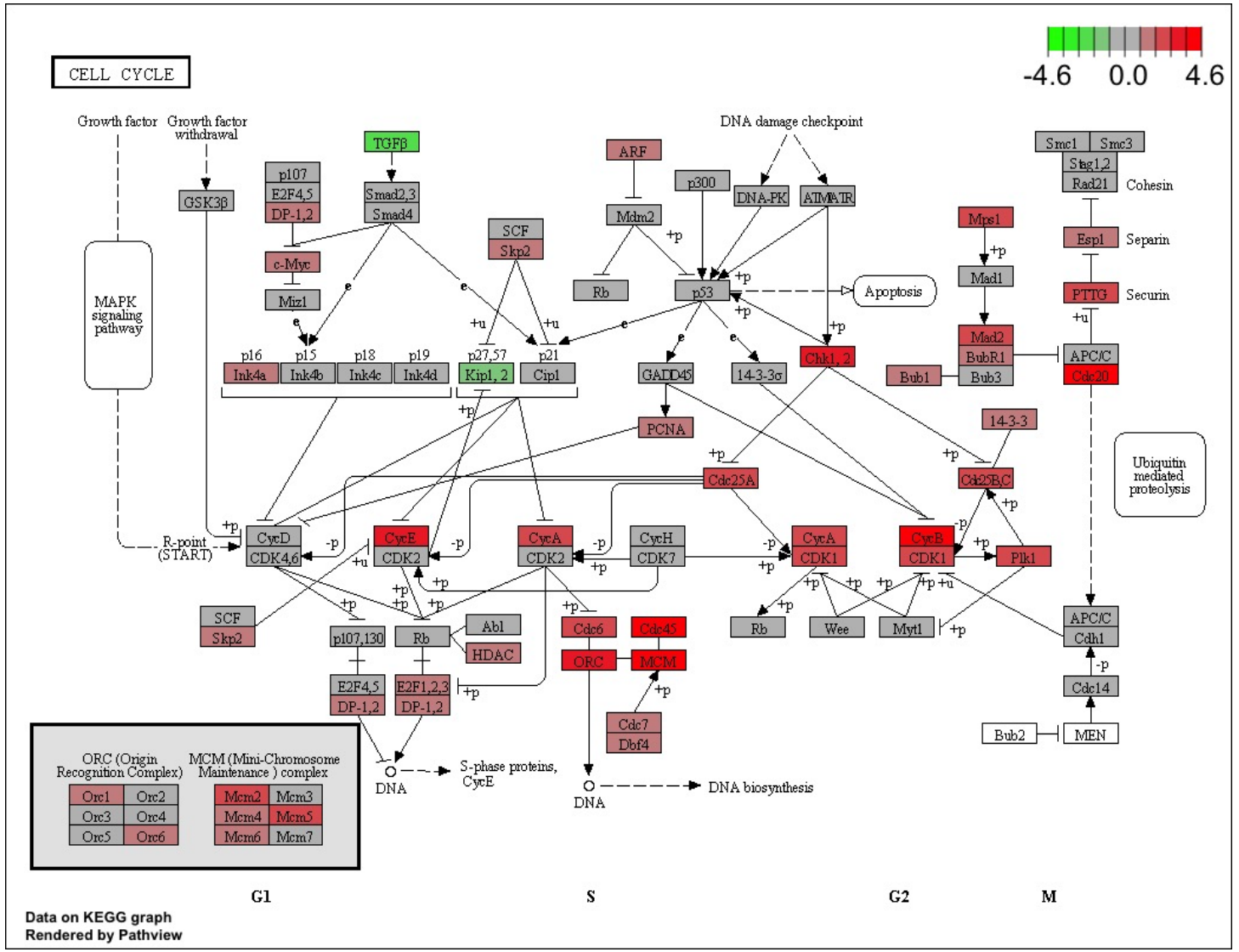


heatplot

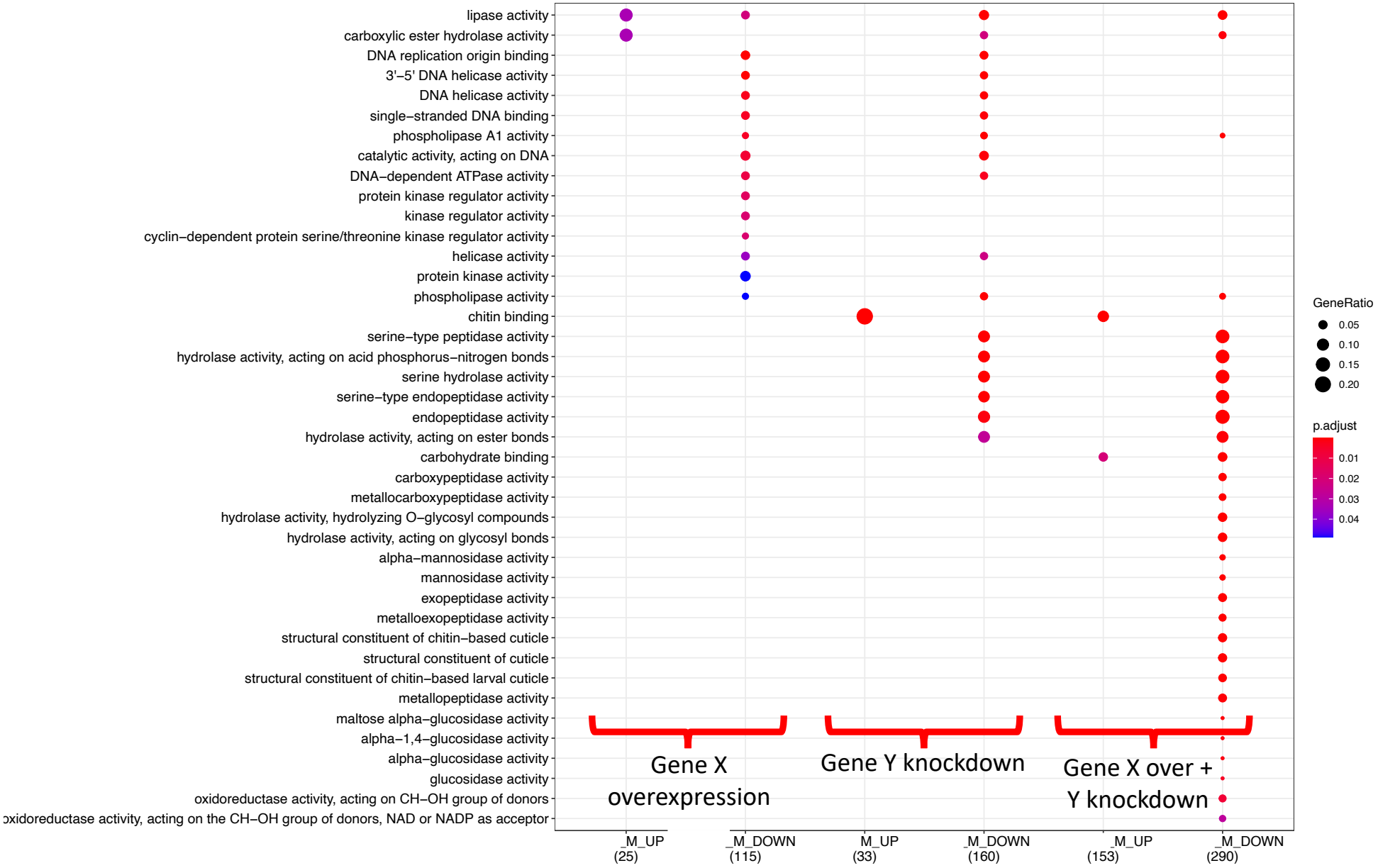


ClusterProfiler – examples of analysis plots

Pathview plots (KEGG-specific)



ClusterProfiler – compare profiles (“Biological theme comparison”)



ClusterProfiler wrapper

- R package wrapper around clusterProfiler for analyzing and visualizing functional profiles (pathways) of gene and gene clusters from an edgeR differential expression table (results from a pairwise-comparison)

- **Supported analyses:**
  - Over-representation analysis
  - Gene set enrichment analysis
  - Biological theme comparison

- **Pathway databases:**
  - Gene Ontology (GO) – cell processes,
  - Kyoto Encyclopedia of Genes and Genomes (KEGG)
  - Reactome
  - MsigDB

} Databases that wrapper calls on so far…

- **Visualization (plot types):**
  - barplot
  - cnetplot
  - dotplot
  - emapplot
  - gseaplot
  - goplot
  - upsetplot
  - heatplot
  - pathview plots (KEGG)
  - ridgeplot

Plots generated will also be appropriately sized!

ClusterProfiler wrapper supported organisms

species_databases.txt

USED BY Msigdb	USED BY GO	USED BY KEGG	USED BY REACTOME
msig_species	db	KEGG_species	reactome_species
Bos taurus	org.Bt.eg.db	bta	
Caenorhabditis elegans	org.Ce.eg.db	cel	celegans
Canis lupus familiaris	org.Cf.eg.db	cfa	canine
Danio rerio	org.Dr.eg.db	dre	zebrafish
Drosophila melanogaster	org.Dm.eg.db	dme	fly
Gallus gallus	org.Gg.eg.db	gga	
Homo sapiens	org.Hs.eg.db	hsa	human
Mus musculus	org.Mm.eg.db	mmu	mouse
Rattus norvegicus	org.Rn.eg.db	rno	rat
Saccharomyces cerevisiae	org.Sc.eg.db	sce	yeast
Sus scrofa	org.Ss.eg.db	ssc	

- 11 organisms so far

.

.

.

Can add additional species to this file, and in theory, the Rscript should work.
AT A MINIMUM, will need information in the first two columns.

ClusterProfiler wrapper supported organisms

species_databases.txt

msig_species	db	KEGG_species	reactome_species
Bos taurus	org.Bt.eg.db	bta	
Caenorhabditis elegans	org.Ce.eg.db	cel	celegans
Canis lupus familiaris	org.Cf.eg.db	cfa	canine
Danio rerio	org.Dr.eg.db	dre	zebrafish
Drosophila melanogaster	org.Dm.eg.db	dme	fly
Gallus gallus	org.Gg.eg.db	gga	
Homo sapiens	org.Hs.eg.db	hsa	human
Mus musculus	org.Mm.eg.db	mmu	mouse
Rattus norvegicus	org.Rn.eg.db	rno	rat
Saccharomyces cerevisiae	org.Sc.eg.db	sce	yeast
Sus scrofa	org.Ss.eg.db	ssc	

This is an actually a Bioconductor package name that is loaded (required for gene ID conversions to ENTREZ IDs)

- 11 organisms so far

Can add additional species to this file, and in theory, the Rscript should work.
AT A MINIMUM, will need information in the first two columns.

ClusterProfiler wrapper usage

1. There are two files you need to have in the same directory:
   - run_clusterProfiler.R
   - species_databases.txt

2. Load R/3.6.0 (TESTED)

3. Run clusterprofiler wrapper:

```
Rscript  run_clusterProfiler.R -f edgeR_DEresults_table.txt -s Mus_musculus
```

Will install all the required packages – including the organism database package (e.g. org.Hs.eg.db)

Required arguments:

-f	DE results table from edgeR (either all genes or the subset of DE genes)
-s	Organism name (Genus_species)

ClusterProfiler wrapper options and default settings

Option	Option Description	Default
-d [string]	Output directory name to save results to (creates a folder called "clusterprofiler_results"	Current working directory (./)
-i [string]	GeneID format/type. Either: ENSEMBL, SYMBOL, ENTREZ, etc. (Can find available gene types from Org.Xx.eg.db)	ENSEMBL
-p [float]	Gene FDR cut-off to apply to the DEG table prior to functional pathway analysis (float) [Use if DEG table contains all significant and non-significant results]	0.05
-c [float]	Log2 fold change cutoff for differentially expressed genes	1
-n [integer]	Number of ontology terms to report (report top X terms)	Reports all terms
-g [integer]	Number of DEGs to use for functional pathway analysis (use top X DEGs) (Genes with p-value less than 0.05 (or user-specified cutoff) are sorted (largest to smallest) by the absolute value of log2FC. Top 200 genes are kept be default).	200
-a [float]	Ontology term over representation test p-value cutoff	0.05
-b [string]	Gene Ontology (GO) domain or branch to query. Either: MF (Molecular Function), CC (Cellular Component), BP (Biological Process), or ALL (all three)	ALL

ClusterProfiler wrapper usage examples

Mouse

```
Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20terms_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20
-p 0.01
```

ClusterProfiler wrapper usage examples

Mouse

Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20term_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20
-p 0.01

Gene IDs are in gene SYMBOL format

gene	logFC	logCPM	PValue	FDR
Alb	1.94993744443276	7.10135330650237	6.36199748453377e-31	9.23189454980696e-27
Serpina1d	2.06959689942489	2.77412789864661	1.67692992276042e-27	1.21669650545882e-23
Serpina3k	1.96134284683396	3.42230224938427	1.61811741853299e-26	6.47867013945331e-23
Apoa2	2.03693652989872	4.66820520379526	1.78586455501435e-26	6.47867013945331e-23
Apob	2.41578948658795	2.13449985687282	2.9510269718012e-23	8.56447047756145e-20
Hamp	2.10889570688613	1.932976995381	1.25588537967055e-21	3.03735879073323e-18
Fgb	2.11803217747982	2.39626624485933	2.90528426897364e-21	6.02265428958236e-18
Serpina1b	1.8313293626374	3.63473198790407	1.94226176981914e-19	3.52302006773069e-16
Arg1	2.15254198817835	1.35628618052214	5.93671673868726e-19	9.37886569681131e-16
Ahsg	2.06155944874045	3.81305181714603	6.46328006120275e-19	9.37886569681131e-16
Ttr	1.9444396721003	4.2712985849673	1.2603610769526e-18	1.66264541705993e-15
Ces1c	2.17094351981178	2.50618326142135	1.74496422779045e-18	2.1100979924556e-15
Mup3	1.9768935351674	3.17220736059439	1.98332849764545e-18	2.21385229456409e-15
Cyp3a11	2.12163599723416	2.08353198479982	8.52514947103537e-18	8.83631742672816e-15

ClusterProfiler wrapper usage examples

Mouse

Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20terms_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20 **[reports top 20 ontology terms – all analyses]**
-p 0.01 **[p-value cut-off for DEGs]**

cnetplot_GO_allterms.pdf	Dec 10, 2020 at 4:34 PM	494 KB	PDF Document
cnetplot_KEGG_allterms.pdf	Dec 10, 2020 at 4:34 PM	24 KB	PDF Document
cnetplot_Reactome_allterms.pdf	Dec 10, 2020 at 4:34 PM	67 KB	PDF Document
Dotplot_GO_over_representation_BP_allterms.pdf	Dec 10, 2020 at 4:34 PM	14 KB	PDF Document
Dotplot_KEGG_over_representation_allterms.pdf	Dec 10, 2020 at 4:34 PM	6 KB	PDF Document
Dotplot_Reactome_over_representation_allterms.pdf	Dec 10, 2020 at 4:34 PM	8 KB	PDF Document
emapplot_GO_BP_allterms.pdf	Dec 10, 2020 at 4:34 PM	2.1 MB	PDF Document
emapplot_KEGG_allterms.pdf	Dec 10, 2020 at 4:34 PM	8 KB	PDF Document
emapplot_Reactome_allterms.pdf	Dec 10, 2020 at 4:34 PM	48 KB	PDF Document
▶ kegg_pathview	Dec 10, 2020 at 4:36 PM	--	Folder
Table_GO_enrichment_results.txt	Dec 10, 2020 at 4:34 PM	26 KB	Plain Text
Table_KEGG_enrichment_results.txt	Dec 10, 2020 at 4:34 PM	1 KB	Plain Text
Table_Reactome_enrichment_results.txt	Dec 10, 2020 at 4:34 PM	4 KB	Plain Text

ClusterProfiler wrapper usage examples

Dog

Rscript run_clusterProfiler.R
-f DE_control-case_list_no_outliers_dog.txt
-d test_dog_top20terms_allGObranch_pval0.01_foldchange1.5x
-s Canis_lupus_familiaris
-n 20 **[reports top 20 ontology terms – all analyses]**
-p 0.01 **[p-value cut-off for DEGs]**
-a 0.01 **[p-value cut-off for Ontology terms]**
-c 1.5 **[genes with log2(FC) > 1.5X will be used for pathway analyses]**

Gene IDs are in ENSEMBL format (which is default setting)

genes	logFC	logCPM	F	PValue	FDR
ENSCAFG00000003154	-2.84382272473412	9.75061879183314	237.273639643645	2.06541472436598e-17	1.73288295374306e-13
ENSCAFG00000010030	-4.74941632710028	8.12279689117867	200.139534674251	2.88755841639442e-16	1.07024427531856e-12
ENSCAFG00000011556	-4.69067052376455	7.33303027353635	196.487221726054	3.82685676514383e-16	1.07024427531856e-12
ENSCAFG00000018268	-3.6977370093103	9.15516301355799	185.922959993012	8.86786626773529e-16	1.7129221101118e-12
ENSCAFG00000004797	-4.85246747736131	5.8722929247539	180.851801867146	1.34671487042285e-15	1.7129221101118e-12
ENSCAFG00000011038	-1.76332135815208	6.34337605232063	180.786511629513	1.35406532179372e-15	1.7129221101118e-12
ENSCAFG00000007669	-1.85262357678111	7.63612341627171	180.140354653326	1.42913644467015e-15	1.7129221101118e-12
ENSCAFG00000011429	-2.9206495650047	7.6651738216944	174.984559719148	2.21117764799405e-15	2.2569971736717e-12
ENSCAFG00000015640	-1.70137567530472	7.83657428648277	173.244550438965	2.56831285330571e-15	2.2569971736717e-12
ENSCAFG00000002318	-3.75494603071666	8.44905571731439	172.608843649505	2.71356472808719e-15	2.2569971736717e-12
ENSCAFG00000009182	-3.80344439012912	10.0153278830763	171.611777070134	2.95911429206063e-15	2.2569971736717e-12
ENSCAFG00000002305	2.24362383083188	3.94731063325776	168.247053507303	3.97627671206948e-15	2.78008013452191e-12

ClusterProfiler wrapper usage examples

Zebrafish

```
Rscript run_clusterProfiler.R
-f Danio_rerio_DE_table_madefromhumanDFV.txt
-d test_drerio_top10terms_top150genes
-s Danio_rerio
-n 10
-g 150
```

[reports top 10 ontology terms – all analyses]

[use top 150 genes for pathway analysis]

ClusterProfiler wrapper uses Over Representation Analysis(ORA) for GO, KEGG, and Reactome

ORA

- Analyses whether a supplied gene (DEG) list is significantly associated with a particular pathway or set of pathways
- Requires a list of gene identifiers (universe) as background
  - This can either be a subset of genes (all genes that are significantly different in the comparison) or all genes
  - The clusterProfiler wrapper uses **all genes** as the universe. Gene information is pulled from the respective organism database.

ClusterProfiler wrapper pros and cons

PROS

- Generates dotplots, cnetplots, emapplots for overrepresented terms from GO, KEGG, and REACTOME pathways
- **All plots that are generated are (hopefully) sized appropriately**
- Saves ontology term results tables
- Runs KEGG pathview on all enriched KEGG terms

CONS

- Does not currently handle a simple list of genes
- Currently does not perform GSEA
- Does not handle Biological theme comparisons

What I would like to include:

- Option to activate pathview for KEGG (it takes a long time to run), default setting = off
- GO term slimming option → refine to most descriptive child terms
- Ability to handle a list of genes
- Molecular Signatures DB?
- Other things?

ClusterProfiler wrapper

Download from Github:

https://github.umn.edu/MSI-RIS/clusterProfiler_wrapper

NEW: Clusterprofiler 1.6 can perform Gene Set Enrichment Analysis (GSEA)!

To enable GSEA, add the "--gsea TRUE" option to your clusterprofiler command line. By default, it will rank all the genes in the DEG table by their log2(fold changes) and test for gene set enrichment based on the position of genes from each hallmark gene set (from the Molecular Signatures Database, hallmark is default) in the ranked list. See the options on the next page if you would like to change the Molecular Signatures Database gene set or the gene ranking method that is used for GSEA.

Important requirements for running GSEA:

- The DEG results table must have been produced with either edgeR, DESeq2, or related method that does proper count normalization.
  - Why? GSEA uses log2(fold change) values to rank genes. Methods like edgeR and DESeq2 perform proper counts normalizations on the samples before calculating gene log2FCs between the treatment groups. Simply taking group count averages and calculating fold change between them is not advised.
- The input DEG results table must have **ALL** of the genes that were tested (not just the significant ones)

ClusterProfiler wrapper v1.6 – new options!

Option	Option Description	Default
--gsea [TRUE/FALSE]	Enable/Disable Gene Set Enrichment Analysis (GSEA).	FALSE
--msigdbs [character]	Molecular Signatures Database for GSEA. Specify a single database, a comma-separated list of databases (no spaces), or “all” to run GSEA against all databases. Available databases: c1, c2_cgp, c2_cp, c2_cp_biocarta, c2_cp_kegg, c2_cp_pid, c2_cp_reactome, c2_cp_wikipathways, c3_mir_mir_legacy, c3_mir_mirdb,c3_tft_gtrd, c3_tft_tft_legacy, c4_cgn, c4_cm, c5_go_bp, c5_go_cc, c5_go_mf, c5_hpo, c6, c7, c8, h. For more information about each, see: http://www.gsea-msigdb.org/gsea/msigdb/index.jsp	h (Hallmark)
--prerankMethod [lfc or pval]	Rank genes based on log2(fold change) or log10(p-value)*direction of fold change for GSEA. If “pval” is specified, but no ”Pvalue” column is available, it will default to using log2FC (lfc).	lfc
--keggpathview [TRUE/FALSE]	Generate KEGG pathview plots for all significant KEGG terms	FALSE

ClusterProfiler wrapper v1.6 - example

The following R command will run the “bos_taurus_DEG_table.txt” DEG table through clusterprofiler using the cow annotations. ORA and GSEA will be run. The default log2FC and FDR cutoffs will be applied to ORA, and all genes will be ranked using log2FC for GSEA. GSEA will be run with the reactome, KEGG, and all GO branchs from the molecular signatures database. Kegg pathviews of ORA KEGG results will not be generated.

```
Rscript   run_clusterProfiler_v1.6.R   -f bos_taurus_DEG_table.txt  -s Bos_taurus   --gsea TRUE   --msigdbs
c2_cp_reactome,c2_cp_kegg,c5_go_bp,c5_go_mf,c5_go_cc --prerankMethod lfc
```

ClusterProfiler wrapper v1.6

Download from Github:

https://github.umn.edu/MSI-RIS/clusterProfiler_wrapper/tree/clusterProfiler_wrapper_v1.6